

1. The real numbers: a brief construction

Dedekind cuts and the number line

Start with the rational numbers $\mathbb{Q}$, their familiar arithmetic, and their order. A **Dedekind cut** is a subset $A \subset \mathbb{Q}$ which is nonempty, is not all of $\mathbb{Q}$, is downward closed, and has no greatest element. Explicitly,

$$r \in A, s < r \implies s \in A; \quad r \in A \implies \exists s \in A : r < s.$$

Think of A as all the rational numbers to the left of a position on the number line. The position need not itself be rational.

Real numbers and rational numbers. Define $\mathbb{R}$ to be the set of all Dedekind cuts. Identify $a \in \mathbb{Q}$ with $a^* = \{r \in \mathbb{Q} : r < a\}$. In particular, $0^* = \{r : r < 0\}$, $1^* = \{r : r < 1\}$. The cut $\{r \in \mathbb{Q} : r < 0 \text{ or } r^2 < 2\}$ is the real number which will be called $\sqrt{2}$.

Order and arithmetic on cuts

All lowercase variables in these formulas range over $\mathbb{Q}$. Define $A \leq B$ by $A \subseteq B$, and $A < B$ by proper inclusion.

$$A + B = \{r + s : r \in A, s \in B\}, \\ -A = \{q : \exists r \notin A \text{ with } q < -r\}.$$

For $A, B \geq 0^*$, define

$$AB = \{q : q < 0\} \cup \{q : \exists r \in A, s \in B, r, s > 0, q < rs\}.$$

Extend multiplication by signs: with exactly one negative factor, negate the product of the positive magnitudes; with two negative factors, multiply their positive magnitudes. A zero factor gives 0^* .

For $A > 0^*$, define

$$A^{-1} = \{q : q \leq 0\} \cup \{q : \exists r > 0, r \notin A, q < 1/r\}.$$

For $A < 0^*$, set $A^{-1} = -((-A)^{-1})$. Finally, $A - B = A + (-B)$ and $A/B = AB^{-1}$ for $B \neq 0^*$.

What the construction gives (without proof). These definitions give $\mathbb{R}$ addition, subtraction, multiplication, division by nonzero numbers, and an order extending those of $\mathbb{Q}$. The familiar rules of arithmetic remain valid. It is *complete*: every nonempty set of real numbers bounded above has a least upper bound, or *supremum*. For a nonempty bounded-above family of cuts $\mathcal{A}$, its supremum is $\bigcup_{A \in \mathcal{A}} A$.

From now on: work with ordinary real numbers and the number line. The definitions below use ordinary arithmetic, order, and completeness.

2. Integer powers, roots, and rational powers

Extend a familiar operation while preserving its basic laws

Positive integers. For $a \in \mathbb{R}$ and $n \geq 1$, define

$$a^1 = a, \quad a^{n+1} = a^n a.$$

Associativity and commutativity give, for positive integers m, n ,

$$a^{m+n} = a^m a^n, \quad (a^m)^n = a^{mn}, \quad (ab)^n = a^n b^n.$$

For $a \neq 0$, define $a^0 = 1$ and $a^{-n} = 1/a^n$. These choices preserve the exponent laws: in particular, $a^n a^{-n} = a^0 = 1$. For nonzero bases, all three laws hold for arbitrary integer exponents.

Positive roots

For $a > 0$ and a positive integer n , there is a unique positive real number ρ with $\rho^n = a$. We denote it by $\sqrt[n]{a}$. We use this familiar fact about real numbers without proof. Existence reflects the absence of gaps in the real line; uniqueness follows because raising a positive number to the n -th power is strictly increasing.

A root is a number; solving an equation is a separate question. $\sqrt{9} = 3$, while $x^2 = 9$ has the two solutions $x = 3$ and $x = -3$. For positive a , $\sqrt[n]{a}$ always denotes the positive root.

Rational exponents | assume $a > 0$ from here onward

For $p \in \mathbb{Z}$ and a positive integer q , define

$$a^{p/q} = (\sqrt[q]{a})^p.$$

Equivalently, $a^{p/q}$ is the unique positive solution of $z^q = a^p$. Thus $a^{p/q} = \sqrt[q]{a^p}$ as well.

Why the fraction representation does not matter. Suppose $p/q = r/s$. Put $u = (\sqrt[q]{a})^p$ and $v = (\sqrt[s]{a})^r$. Then

$$u^{qs} = a^{ps} = a^{rq} = v^{qs}.$$

Since $u, v > 0$, uniqueness of the positive qs -th root gives $u = v$. This is why $a^{1/2}$ and $a^{2/4}$ denote the same number.

Putting exponents over a common denominator gives, for $r, s \in \mathbb{Q}$ and $a, b > 0$,

$$a^{r+s} = a^r a^s, \quad (a^r)^s = a^{rs}, \quad (ab)^r = a^r b^r.$$

For example, $16^{3/4} = 8$, $8^{-2/3} = 1/4$, and $a^{1/2} a^{1/3} = a^{5/6}$.

3. Real exponents

Use order to fill in the remaining exponents

The expression $2^{3/2}$ now has a meaning. What should $2^{\sqrt{2}}$ mean? For a base greater than one, the values should increase with the exponent. We use rational exponents to locate the required value on the real line.

Recall the supremum. A number M is an upper bound of S if $s \leq M$ for every $s \in S$. The supremum $\sup S$ is its least upper bound. It need not be in the set: $\sup(0, 1) = 1$.

Definition of powers with real exponents

For $a > 1$ and $x \in \mathbb{R}$, define

$$a^x = \sup\{a^r : r \in \mathbb{Q}, r \leq x\}.$$

For the remaining positive bases, set

$$1^x = 1, \quad a^x = \frac{1}{(1/a)^x} \quad \text{when } 0 < a < 1.$$

Why this defines a number. Choose integers $m \leq x < M$. For $a > 1$, rational powers are strictly increasing: if $r < s$, then $a^s/a^r = a^{s-r} > 1$. Thus a^m belongs to the defining set and a^M bounds it above. Completeness supplies its supremum, a positive real number.

Agreement with rational powers. If $x \in \mathbb{Q}$, the old value a^x is the largest member of the defining set, hence its supremum. For $0 < a < 1$, agreement follows from $(1/a)^r = 1/a^r$.

Order. If $a > 1$ and $x < y$, choose rationals r, s with $x < r < s < y$. Then

$$a^x \leq a^r < a^s \leq a^y.$$

So a^x is strictly increasing. Reciprocals show it is strictly decreasing for $0 < a < 1$.

An irrational power between familiar powers. Since $(7/5)^2 < 2 < (3/2)^2$,

$$2^{7/5} < 2^{\sqrt{2}} < 2^{3/2} = 2\sqrt{2}.$$

Closer rational bounds locate the irrational power more precisely. This locates the value using the order of the real numbers.

Why the definition depends on the base

If $0 < a < 1$, the numbers $a^{-n} = (1/a)^n$ become arbitrarily large. All sufficiently negative integers lie below any fixed x . Thus $\{a^r : r \in \mathbb{Q}, r \leq x\}$ is unbounded above. Defining a^x by reciprocals avoids this problem.

4. Exponential functions

Algebraic laws and the shape of the graph

Properties of real powers

For $a, b > 0$ and $x, y \in \mathbb{R}$,

$$a^{x+y} = a^x a^y, \quad a^{x-y} = \frac{a^x}{a^y}, \quad (a^x)^y = a^{xy}, \quad (ab)^x = a^x b^x.$$

The exponential function $x \mapsto a^x$ has domain $\mathbb{R}$. For $a \neq 1$, its range is $(0, \infty)$; for $a = 1$, its range is $\{1\}$.

The extension from rational to real exponents preserves these laws and supplies every positive output when $a \neq 1$. These facts are stated here without proof. Intuitively, the exponential graph has no gaps. All its values are positive and it passes through $(0, 1)$. The identity $(1/a)^x = a^{-x}$ reflects the graph across the y -axis.

Worked example: Algebra with irrational exponents

For $a > 0$,

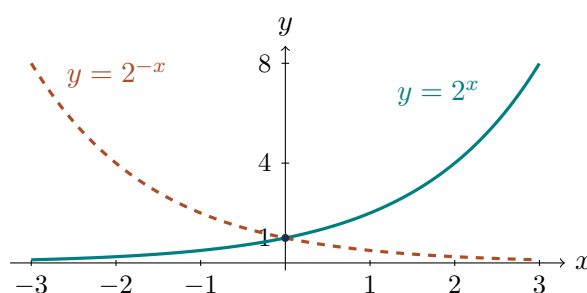
$$\frac{a^{\sqrt{2}+1}}{a^{\sqrt{2}-1}} = a^2, \quad (a^{\sqrt{2}})^{\sqrt{2}} = a^2.$$

If $a \neq 1$, strict monotonicity gives $a^u = a^v \iff u = v$. Thus $2^{x+1} = 8^{x-1}$ is equivalent to $x+1 = 3x-3$, giving $x = 2$.

Why positive bases? Negative bases have some real powers, but $(-1)^{1/2}$ is not real. Even when both expressions exist, the unrestricted power law can fail:

$$((-1)^2)^{1/2} = 1 \neq -1 = (-1)^{2(1/2)}.$$

For zero, negative powers would require division by zero. We leave 0^0 undefined in this lecture.



Reading. Stewart, *Calculus: Early Transcendentals*, 9th ed., Section 1.4, pp. 45–51. The construction of real numbers and the systematic extension of powers supplement the textbook.